

# Inference as Training

## *A Residue-and-Reset Criterion for State-Changing Inference*

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### **Abstract**

Suppose a person misjudges the weight of a cup, corrects the movement, and lifts it more accurately on the next attempt. The later success may reflect more computation with an unchanged system, or it may reflect a change left by the first attempt. Those possibilities are often blurred when inference, adaptation, memory, and learning are discussed across neuroscience and artificial intelligence.

This paper supplies an operational boundary. An inference episode becomes training-like when mismatch resolution changes a measurable state, that state persists beyond the immediate episode, and selectively removing it changes a later inference under matched conditions. The retained causal state is called **residue**. The framework separates fast context or routing updates from slower consolidation, distinguishes output change from performance improvement, and makes maladaptive learning visible: a retained state can satisfy the training-like criterion while worsening later performance.

Fourteen numbered equations formalize the inference state, signed mismatch, update gate, fast and slow updates, reset intervention, residue magnitude, controlled future effect, performance effect, compute-matching requirement, persistence profile, classification rule, stability-plasticity harm, and a benchmark data model. A deterministic twelve-seed scalar-shift benchmark then tests the logic. Mean later-query root-mean-square error is 1.5003 with no update, 0.1215 with a retained gated update, 1.5003 after the same update computation followed by reset, and 2.9943 with a sign-flipped wrong-context residue. A below-threshold null condition keeps the gate closed, retains zero state, and has zero future output effect. After the shift reverses, overpersistent carry produces error 2.9988 versus 1.5049 after reset; a fresh gated update reduces it to 0.1222. Normal and optimized Python replays are byte-identical, and all prespecified per-seed checks pass.

The benchmark is a unit test of the discriminator, not evidence for a universal biological mechanism. Biological plasticity, AI test-time adaptation, and thermodynamic mismatch reduction remain separate evidence classes. The contribution is a common causal question that can be asked of each: what changed, how long did it remain, and does resetting that change remove its later effect?

## **Scope and evidence boundary**

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## **1. The Ordinary Problem: Why Was the Second Attempt Different?**

Imagine lifting a cup that is much lighter than expected. The first movement overshoots. Sensory feedback arrives, the movement is corrected, and the next lift is better calibrated. Several different things may have happened:

1. The nervous system may simply have continued the same inference for longer.
2. A temporary state may have remained active until the second movement.
3. A route, gain, or short-lived synaptic variable may have changed.

4. A durable memory or plastic change may have been written.
5. The first correction may have produced a change that is useful in the original setting but harmful after the cup is replaced.

All five can look like “the system learned” if only the second performance is measured. They are not the same mechanism.

The unexplained problem is therefore not whether a system changed somewhere. Any physical system changes while operating. The problem is to identify a state change that crosses the boundary between one inference episode and a later one, and then to show that the retained change causes the later difference.

The same problem appears in artificial systems. A model can answer differently because it received more tokens, because a recurrent state remained active, because a memory was written and retrieved, because normalization statistics or parameters changed at test time, or because the entire evaluation was rerun with more compute. These possibilities cannot be distinguished by a performance score alone.

The central operation can be stated without specialized vocabulary: a system encounters a discrepancy, decides whether that discrepancy is reliable enough to act on, changes a fast or slow internal variable, carries some of that change forward, and behaves differently later because the carried variable remains available. Before naming that operation, the next section distinguishes it from the established accounts that already explain prediction, current-state correction, fast memory, test-time adaptation, and plasticity.

## 2. What Established Accounts Already Explain

Several established literatures supply pieces of this sequence.

Predictive-coding models distinguish top-down predictions from residual signals that update current representations [1]. The free-energy framework distinguishes changes in activity during perceptual inference from slower changes in synaptic efficacy during learning, as well as changes in gain associated with attention [2]. Those distinctions already warn against treating every error-sensitive activation as learning.

Artificial-intelligence research makes update surfaces unusually visible. Test-Time Training converts an unlabeled test sample into a self-supervised problem and updates model parameters before prediction [3]. TENT adapts normalization statistics and affine parameters online by entropy minimization [4]. Fast-weight systems introduce a timescale between momentary activity and slow standard weights [5]. These are concrete update-at-inference mechanisms, but they do not by themselves establish a biological mechanism.

Neuroscience also supplies multiple candidate residue timescales. Residual presynaptic calcium has been proposed as a short-lived working-memory buffer [6]. Spike timing can determine the sign and magnitude of persistent synaptic modification [7]. Behavioral time-scale synaptic plasticity can associate seconds-long behavioral episodes with subsequent place-field formation [8]. Homeostatic plasticity supplies stabilizing counterforces against runaway excitation or unbounded weight change [9]. KIBRA / PKM zeta and CaMKII studies address molecular or synaptic persistence despite turnover [10,11].

Research on biologically plausible credit assignment further shows why “feedback-driven plasticity” is a more precise family name than a blanket identity with ordinary backpropagation. Random feedback can support learning in artificial networks [12], and predictive-coding networks can approximate backpropagation under specified assumptions using local Hebbian plasticity [13]. Prospective-coding models show how temporally local plasticity can learn future-oriented responses over longer behavioral intervals [14]. These are specific models with conditions, not permission to collapse all biological feedback into one algorithm.

What is missing across these literatures is a compact cross-domain discriminator. Current activity, fast state, memory, parameters, molecular traces, and structural plasticity all differ, but the same controlled question can be asked: does a measurable after-effect survive, and does removing it change later inference?

### 3. The Plain-Language Operation

The proposed measurement sequence has six steps.

1. **Produce an inference.** The system estimates a sensory cause, chooses an action, generates an answer, or predicts an outcome.
2. **Measure a discrepancy.** A target, observation, outcome, consistency check, or self-supervised objective differs from the current estimate.
3. **Apply an update gate.** The system decides whether the discrepancy is large, reliable, relevant, and permitted to alter state.
4. **Change one or more variables.** A fast context, phase, routing, gain, memory, normalization statistic, adapter, or parameter changes. Repeated or important fast changes may be consolidated into a slower variable.
5. **Wait until a later query.** Immediate activity is allowed to end. The experiment identifies what, if anything, remains.
6. **Reset only the candidate change.** The later query is repeated with the candidate retained state selectively removed while input, compute, and other relevant conditions are matched.

The surviving state in step 5 is the **residue**. The change in later output caused by step 6 is the **controlled future effect**. The reset is essential because a later improvement can otherwise be attributed to more computation, repeated exposure, longer context, or a different query.

Residue is a measurement role, not a claim that every domain shares one physical substrate. A written memory object, a fast weight, residual presynaptic calcium, a routing bias, and a long-term synaptic change may all satisfy the role while remaining mechanistically distinct.

The ordinary problem, inherited accounts, and future-effect operation are now in place. **Inference as training** names the subset of inference episodes in which mismatch resolution leaves a measurable residue and selective reset of that residue removes a later effect. The term summarizes the operation; it does not substitute for it or rename ordinary fixed-state evaluation.

### 4. Formal Measurement Contract

#### 4.1 State and current inference

Let  $x_t$  be the current input;  $c_t$  the measured context;  $s_t$  a fast state; and  $\theta_t$  a slower state or parameter set. The current output is

$$\hat{y}_t = F(x_t; \theta_t, s_t, c_t). \quad (1)$$

The separation between  $s_t$  and  $\theta_t$  is functional rather than absolute. A variable is fast or slow relative to the episode and query interval being tested.

## 4.2 Signed mismatch and mismatch magnitude

Let  $y_t^*$  denote an observed outcome, labeled target, sensory return, intended result, or internally defined consistency reference. A signed discrepancy  $r_t$  preserves update direction, while  $e_t$  measures its magnitude:

$$\begin{aligned} r_t &= y_t^* - \hat{y}_t, \\ e_t &= d_t(r_t, 0). \end{aligned} \tag{2}$$

For unlabeled test-time adaptation,  $y_t^*$  can be replaced by an explicitly named proxy objective. The proxy must not be described as ground truth merely because it drives an update.

## 4.3 Update gate

Not every discrepancy should alter state. Let  $a_t$  encode update availability and let  $q_t$  encode a relevance or quality test. One simple gate is

$$g_t = \mathbf{1}[e_t \geq \tau_e] \mathbf{1}[q_t \geq \tau_q] a_t, \quad g_t \in \{0, 1\}. \tag{3}$$

The gate makes attention, confidence, novelty, task relevance, neuromodulation, or policy permission measurable instead of treating every activation as an update.

## 4.4 Fast update

A fast state can decay, integrate discrepancy, and alter subsequent processing:

$$s_{t+1} = (1 - \lambda_s) s_t + \eta_s g_t B_t(r_t, x_t, c_t), \quad 0 \leq \lambda_s \leq 1. \tag{4}$$

Examples include working context, routing state, short-term facilitation, temporary weights, normalization statistics, or a phase / timing relation. The map  $B_t$  must be specified for a concrete implementation.

## 4.5 Slow update and consolidation

Let  $h_t \in [0, 1]$  be a consolidation gate based on repetition, salience, verification, reward, or another measured condition. A slower update is

$$\theta_{t+1} = \theta_t + \eta_\theta h_t C_t(s_{t+1}, r_t, x_t, c_t). \tag{5}$$

Equations (4) and (5) express the Draft 6 fast / slow distinction without implying that every fast update must become structural. Consolidation is a separate operation with separate failure modes.

## 4.6 Selective reset

Let  $z_t = (s_t, \theta_t)$  collect the candidate state dimensions. A reset operator removes only the update made available by the episode while preserving fixed architecture and matched nuisance variables:

$$Q_t(z_{t+k}^{\text{retain}}) = z_{t+k}^{\text{reset}}. \tag{6}$$

The exact intervention depends on the domain. It may clear a recurrent state, disable retrieval, restore a checkpoint, zero a fast adapter, reverse a pharmacological manipulation, or compare trials in which a plasticity marker is selectively blocked.

#### 4.7 Residue magnitude

The candidate residue at lag  $k > 0$  is

$$\rho_t(k) = d_z\left(z_{t+k}^{\text{retain}}, Q_t(z_{t+k}^{\text{retain}})\right). \quad (7)$$

A nonzero  $\rho_t(k)$  proves only that a state survived. It does not yet prove that the state mattered to later inference.

#### 4.8 Controlled future output effect

Let  $\mathcal{D}_{t+k}$  be a matched later-query distribution. The causal output effect of retaining the candidate state is

$$\Delta_y(t, k) = \mathbb{E}_{x \sim \mathcal{D}_{t+k}} \left[ d_y\left(F(x; z_{t+k}^{\text{retain}}), F(x; Q_t(z_{t+k}^{\text{retain}}))\right) \right]. \quad (8)$$

This is the decisive dependency test. If reset removes no later output difference, the measured state is not the operative residue for that query.

#### 4.9 Benefit is separate from causal effect

When a target or utility measure exists, define the retained-state benefit as

$$\Delta_{\mathcal{L}}(t, k) = \mathbb{E} \left[ \mathcal{L}\left(F(x; Q_t(z_{t+k}^{\text{retain}})), y^*\right) - \mathcal{L}\left(F(x; z_{t+k}^{\text{retain}}), y^*\right) \right]. \quad (9)$$

A positive value means retention helps; zero means no measured performance effect; a negative value means the retained update is harmful. The expectation is over the same matched later-query distribution used for Equation (8). Training-like state change and successful learning are therefore not synonyms.

#### 4.10 Compute-matched intervention

To isolate residue from extra effort, the retained and reset arms must satisfy

$$\begin{aligned} \mathcal{C}_{\text{retain}} &= \mathcal{C}_{\text{reset}}, \\ \mathcal{I}_{\text{retain}} &= \mathcal{I}_{\text{reset}}, \end{aligned} \quad (10)$$

where  $\mathcal{C}$  denotes the prespecified update computation and  $\mathcal{I}$  denotes matched later-query inputs. The reset arm performs the update computation and then removes its candidate state before the later query.

### 4.11 Persistence profile

Persistence can be expressed relative to the earliest post-episode residue:

$$\begin{aligned} P_t(k) &= \frac{\rho_t(k)}{\rho_t(k_0)}, \\ k_{1/2} &= \inf \left\{ k \geq k_0 : P_t(k) \leq \frac{1}{2} \right\}, \\ \rho_t(k_0) &> 0. \end{aligned} \tag{11}$$

This allows immediate, short-persistence, and structural residues to be compared without pretending that they share a timescale. When  $\rho_t(k_0) = 0$ , the normalized persistence profile and half-life are undefined rather than zero: there is no detected reference residue to normalize, and that episode fails the residue limb of the training-like criterion.

### 4.12 Operational classification

For tolerances  $\delta_z > 0$  and  $\delta_y > 0$ , an episode is classified as training-like at lag  $k$  when

$$T_t(k) = \mathbf{1} [g_t = 1 \wedge \rho_t(k) > \delta_z \wedge \Delta_y(t, k) > \delta_y]. \tag{12}$$

This rule does not require  $\Delta_{\mathcal{L}} > 0$ . It identifies state-changing inference; performance and safety are evaluated separately.

### 4.13 Stability-plasticity harm under reversal

If the environment changes from distribution  $\mathcal{D}_A$  to  $\mathcal{D}_B$ , overpersistent carry can be measured by

$$H_{A \rightarrow B} = \mathbb{E}_{(x,y) \sim \mathcal{D}_B} [\mathcal{L}(F(x; z_A^{\text{carry}}), y) - \mathcal{L}(F(x; \mathcal{Q}(z_A^{\text{carry}})), y)]. \tag{13}$$

A positive  $H_{A \rightarrow B}$  means that old residue harms performance after reversal. This makes forgetting, reset, context separation, and homeostatic control part of the theory rather than afterthoughts.

## 5. Residue Classes and Their Necessary Controls

Training-like inference can operate through several residue classes. They should be compared by role while remaining distinct by mechanism.

| Residue class           | What changes                                                     | Later-query measurement                             | Necessary control                                                     |
|-------------------------|------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------|
| Context residue         | Prompt, scratchpad, critique, plan, or active task state         | Later output depends on retained structured context | Same context budget with the structured residue removed or randomized |
| Routing residue         | Module choice, retrieval path, policy branch, or attention route | Later path selection changes selectively            | Forced original route or routing reset                                |
| Recurrent-state residue | Hidden state carried across steps or episodes                    | State reset removes later difference                | Reset-state and same-compute control                                  |

| Residue class            | What changes                                                           | Later-query measurement                         | Necessary control                                          |
|--------------------------|------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------|
| Temporary-weight residue | Fast weight, adapter, normalization statistic, or local parameter      | Same-shift later samples change                 | No-update and update-then-reset baselines                  |
| Memory residue           | Durable note, vector, trace, or object                                 | Later inference retrieves and uses the object   | Retrieval disabled, stale memory, and irrelevant memory    |
| Biological gain residue  | Excitability, response probability, gain, or timing                    | Later neural response changes                   | Arousal, attention, movement, and sensory-history controls |
| Plasticity residue       | LTP / LTD, BTSP, synaptic efficacy, molecular tag, or structural trace | Marker or perturbation predicts later response  | Timing-window and marker-specific intervention             |
| Phase / timing residue   | Relative phase, settling time, coupling, or temporal route             | Later coordination or receiver response changes | Phase scramble, sham timing, or selective reset            |

The strongest causal evidence uses a selective intervention. A correlation between a marker and later behavior is weaker than showing that the later effect disappears when the marker or access route is removed.

## 6. Five Operational Timescales, Not One

The paper preserves the Draft 6 distinction among immediate, short, and structural change.

| Scale                         | Representative variables                                                                                      | Classification                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Instantaneous operation       | A spike, activation, token step, or local phase event                                                         | Current computation only                                                 |
| Within-episode settling       | Attention, iterative error correction, recurrent settling, or phase realignment during one bounded inference  | Ordinary inference unless the state survives the episode boundary        |
| Cross-episode working residue | Scratchpad, recurrent state, residual calcium, eligibility trace, or active task context                      | Training-like when reset removes a later-query effect                    |
| Intermediate adaptation       | Fast weight, adapter, normalization statistic, routing bias, or repeated local update                         | Training-like when retained state remains causal across the stated lag   |
| Structural persistence        | Durable memory, standard weights, synaptic efficacy, long-term routing, molecular or anatomical consolidation | Strong persistence case, but still requires a reset or perturbation test |

Fast and slow are relational terms. A one-second synaptic variable can be slow relative to a millisecond spike and fast relative to a day-long memory. The experiment must name the episode boundary and query lag before assigning a residue class.

The two-timescale formulation also exposes a design problem. If fast state never consolidates, useful corrections disappear. If every fast discrepancy consolidates, noise and wrong-context information accumulate. Equation (5) therefore includes a consolidation gate rather than making persistence automatic.

## 7. Biological Interpretation and the SAN Genealogy

### 7.1 Bounded public development ladder, 2012-2022

The present discriminator did not appear all at once. Earlier public records contain partial ancestors, and their date types must remain separate from the mature formulation. The table therefore reports only public

video upload dates or immutable Git fixation dates. It does not convert later terminology into wording that the earlier sources did not contain.

| Public fixation                                                      | Bounded content present in that source                                                                                                                                                                                                                                                                        | Relationship to the present paper                                                                                                                             |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 31 and August 23, 2012, public video uploads [19,20]            | EEG-linked sensory feedback is described as a closed cycle in which measured brain activity changes light or sound, the changed stimulus returns through the senses, and temporally aligned events can be compared. The second dialogue also invokes coactivity as a linkage principle.                       | Public feedback and temporal-coactivity ancestry. These sources do not contain PWD, a formal residue variable, or a reset intervention.                       |
| April 11 and December 21, 2017, public audio / video records [21,22] | The Neural Lace proposal asks for the difference between the pattern currently present and the pattern desired, then proposes sending only that difference. The same program contrasts verified feedback-loop communication with one-way transmission and treats recurrent return as operationally important. | Public current-versus-target and feedback-verification ancestry. It does not yet distinguish retained state from matched extra computation.                   |
| July 5, 2022, immutable Git fixation [23]                            | Phase intervals are proposed to modify a represented pattern as signals traverse successive neurons; action potentials are said to trigger shifts in memory states across the brain.                                                                                                                          | Public iterative-update and memory-state ancestry. Persistence and reset are not operationalized.                                                             |
| August 5, 2022, immutable Git fixation [24]                          | The SAN whitepaper describes memories scaling from synaptic connections through recurrent oscillatory activity, with tonic expectations modified by phasic events and subsequent neural-array transmission.                                                                                                   | Public multiscale memory-update and oscillating-feedback ancestry. The account is conjectural and does not establish a causal reset test.                     |
| September 17, 2022, immutable Git fixation [25]                      | Tonic firing is proposed to maintain active memory state; unusual phase changes are described as persistent enough to alter what downstream arrays receive; synaptic frequencies and pathways are proposed to change across repeated loops.                                                                   | Public state-maintenance, persistence, and receiver-update ancestry. It does not supply the normalized persistence profile or compute-matched counterfactual. |
| November 28, 2022, immutable Git fixation [26]                       | Distinct rendered patterns are described as persisting in cortical, thalamic, and hippocampal feedback loops while changing pathways alter future flow.                                                                                                                                                       | Public loop-carried-pattern ancestry. It remains a qualitative mechanism proposal rather than the present discriminator.                                      |

The ladder supports a bounded genealogy: feedback, temporal alignment, current-versus-target difference, iterative update, active memory, and loop-carried persistence appear in public stages before the mature criterion. It does **not** backdate the 2026 proposition that training-like inference requires a measurable residue whose selective reset removes a matched later effect.

## 7.2 A public 2024 joined ancestor

An immutable September 6, 2024 SAN Git source provides an earlier first-party mechanism ancestor [15]. In Micah Blumberg’s authored portions, an unexpected out-of-sync excitation or inhibition is described as a Phase Wave Differential (PWD); the event is proposed to change synaptic frequency and connectivity, be held and broadcast as an updated rendering, differ from a prior tonic state, and be absorbed by an oscillating group as the difference between successive memory-renderings shrinks.



That 2024 source establishes a joined conjectural chain:

unexpected receiver-relative difference  
-> changed synaptic or oscillatory state  
-> held and broadcast update  
-> reduction or absorption across later activity

It does not already contain the 2026 residue / reset causal discriminator. The present paper turns the older chain into an intervention-capable contract: identify the changed variable, measure its survival, reset it, and test the later effect.

### 7.3 The 2026 public fast / slow bridge

The publicly fixed 2026 prospective-learning paper develops phase frustration as an error-like signal, rapid phase-based adaptation, and slower structural consolidation when a phase-lock signature persists or repeats [16]. That supplies the fast / slow architecture preserved in Equations (4) and (5). An owner-held December 2025 development date is preserved in the private provenance record, but it is not represented here as public fixation. *Building Sentient Beings* separately develops memory-writing and self-updating agent architecture [17].

These first-party sources are development genealogy. They do not replace experimental neuroscience. Their role is to show how the present criterion grows from the author's earlier PWD, prospective-state, and self-updating-agent program.

### 7.4 What PWD adds to inherited vocabulary

Phase coding describes information associated with timing relative to an oscillation. Traveling-wave models describe organized propagation across tissue. Predictive coding describes relations between predictions and residual error. The PWD proposal assigns a more specific operational role: a receiver-relative departure from an expected tonic or routed relation that can change the receiver, alter subsequent transmission, and potentially leave fast or slow residue.

That distinction matters because propagation alone does not establish learning, and error magnitude alone does not establish persistence. A PWD candidate enters the present framework only when its downstream state change is measured and the later effect survives the reset test. This provides a way to test the proposed operation rather than treating a new term as proof.

### 7.5 Biological evidence is scale-specific

The established studies support components, not a single universal chain:

- Bi and Poo show that relative spike timing can determine persistent potentiation or depression under specified culture conditions [7].
- Mongillo, Barak, and Tsodyks propose a residual-calcium buffer whose state can be loaded, refreshed, and read out [6].
- Bittner and colleagues demonstrate a seconds-long plasticity window associated with CA1 place-field formation [8].
- Turrigiano describes homeostatic mechanisms that stabilize neuronal function [9].
- Lee, Chen, and Nicoll show persistence of synaptic memory despite molecular turnover, while Tsokas and colleagues examine KIBRA / PKM zeta anchoring in memory persistence [10,11].
- Predictive-coding and prospective-coding models supply explicit candidate mappings from discrepancy to state change [1,13,14].

The scientific next step is not to declare these mechanisms identical. It is to apply the same residue / reset questions at each scale and determine where the proposed chain actually holds.

## 7.6 Feedback-driven plasticity and the backpropagation boundary

Backpropagation has a specific computational meaning: an objective gradient is transported through an artificial network and used for credit assignment. Biological feedback, phase-sensitive update, predictive-coding relaxation, neuromodulatory gating, and local plasticity do not become that algorithm merely because they are error-sensitive.

The biological mechanism family in this paper is therefore **feedback-driven plasticity**. Feedback alignment [12] and predictive-coding approximations [13] show that exact symmetric transport is not the only route to useful error-sensitive learning. They do not establish that ordinary cortex implements standard artificial-neural-network backpropagation. A biological experiment must name the local variables, update rule, timing window, and causal residue rather than rely on the phrase “biological backpropagation.”

## 8. AI Interpretation

AI systems expose update surfaces that can often be checkpointed and reset exactly.

| System type                   | Candidate state                                | Training-like test                                                                       |
|-------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------|
| Fixed feed-forward evaluation | Current activations only                       | Later query is unchanged after the current episode                                       |
| Context-only adaptation       | Prompt, scratchpad, critique, or plan          | Remove structured context while matching context length and compute                      |
| Retrieval-memory agent        | Written object and retrieval index             | Disable or substitute retrieval on a later matched query                                 |
| Recurrent system              | Hidden state                                   | Reset hidden state while holding model and query fixed                                   |
| Fast-weight system            | Intermediate-timescale weights                 | Restore the pre-episode fast weights [5]                                                 |
| Test-Time Training            | Parameters updated on the test sample          | Restore the pre-update checkpoint [3]                                                    |
| TENT-style adaptation         | Normalization statistics and affine parameters | Restore those statistics and parameters [4]                                              |
| Critique or error-replay loop | Error object plus revised state                | Remove the retained error-derived object while matching extra compute                    |
| SVI-style error bank          | Stored error artifacts across generation       | Engineering comparison owned by the separate oscillatory-state-persistence analysis [18] |

The category “training-like” is intentionally broader than gradient descent. A memory write can change later inference without modifying model weights. A recurrent state can act as a short residue. A parameter update may be training-like even if it later harms performance. What joins the cases is the causal dependency, not one update algorithm.

Three AI controls are especially important:

1. **No-update:** run the same later query without applying the mismatch update.
2. **Compute-matched update-then-reset:** perform the update computation, then restore the candidate state before the later query.
3. **Wrong-context residue:** retain an update derived from an incompatible context or reverse its sign.

The first shows whether adaptation matters. The second distinguishes state retention from extra computation. The third tests whether the residue is content-specific rather than a generic benefit from additional processing.

## 9. Deterministic Residue / Reset Benchmark

### 9.1 Purpose

The benchmark is deliberately small. It is a unit test for the logic of the criterion, not a competitive machine-learning result and not a model of neural plasticity.

Each sample follows a shifted scalar relation:

$$\begin{aligned} y_i &= x_i + b + \varepsilon_i, \\ \varepsilon_i &\sim \mathcal{N}(0, \sigma^2). \end{aligned} \tag{14}$$

The fixed predictor is  $\hat{y}_i = x_i + r$ , where  $r = 0$  initially. A calibration set supplies the signed mean mismatch. The update gate opens when its magnitude exceeds a fixed threshold, and the retained arm writes the mismatch into  $r$ . All arms use identical later-query samples.

### 9.2 Prespecified conditions

- **Fixed/no update:** retain  $r = 0$ .
- **Gated retained update:** compute the calibration mismatch and retain the resulting  $r$ .
- **Compute-matched update-then-reset:** perform the same update computation, then restore  $r = 0$ .
- **Wrong-context / sign-flipped residue:** apply the opposite sign of the retained update to the same-shift query.
- **Overpersistent carry across reversal:** retain the positive-shift residue after the environment changes to the equal negative shift.
- **Fresh gated update after reversal:** reset and recalibrate on the reversed shift.
- **Below-threshold null:** use zero shift with noise small enough that the prespecified mismatch gate remains closed.

The fixed configuration uses twelve named seeds, 32 calibration samples and 256 later-query samples per seed,  $b_A = 1.5$ ,  $b_B = -1.5$ ,  $b_0 = 0$ ,  $\sigma = 0.12$ , learning rate 1.0, and gate threshold 0.25. The implementation uses only the Python standard library.

### 9.3 Results

| Condition                            | Mean later-query RMSE | Interpretation                                                        |
|--------------------------------------|-----------------------|-----------------------------------------------------------------------|
| Fixed/no update                      | 1.5003                | Baseline shift error                                                  |
| Gated retained update                | 0.1215                | Retained state fits the same shift                                    |
| Compute-matched update-then-reset    | 1.5003                | Same update computation without residue reproduces baseline           |
| Wrong-context / sign-flipped residue | 2.9943                | Retained state is content-sensitive and can be harmful                |
| Reversal after reset                 | 1.5049                | Baseline on reversed shift                                            |
| Overpersistent carry across reversal | 2.9988                | Old residue harms after context reversal                              |
| Fresh gated update after reversal    | 0.1222                | New context-specific update recovers                                  |
| Below-threshold null, fixed or gated | 0.1195                | Gate remains closed; retained residue and future effect are both zero |

The retained-state root-mean-square output effect versus reset is 1.4964. Same-shift retained benefit versus reset is 1.3788 RMSE units. Wrong-context harm versus reset is 1.4940, and overpersistent reversal harm

versus reset is 1.4939.

All twelve prespecified aggregate checks pass for every seed:

1. every initial-shift gate opens;
2. every reversal-shift gate opens;
3. retained update beats fixed inference;
4. update-then-reset equals fixed inference within  $10^{-12}$ ;
5. wrong-context residue harms performance;
6. retained state changes later predictions;
7. overpersistent carry harms after reversal;
8. fresh reversal update beats reset;
9. every null-condition gate remains closed;
10. every null-condition residue remains zero;
11. every null-condition future effect remains zero; and
12. null-condition gated error equals fixed error.

The normal and optimized Python result files are byte-identical at SHA-256 101A1D70ACAFB859E4AF19C3960EC650B2FED5EE402F80A5387C8C9D039FE73C. Fourteen deterministic unit tests pass under both interpreters.

## 9.4 What the benchmark establishes

The benchmark establishes that the measurement contract is internally discriminating:

- improvement disappears when the candidate residue is reset despite matched update computation;
- below-threshold mismatch leaves the gate closed and produces no residue or future effect;
- a residue can causally change later inference without being beneficial;
- persistence becomes a liability after reversal; and
- fresh context-specific updating can recover from that liability.

It does not establish that the scalar update is novel, biologically implemented, or sufficient for general intelligence. Its value is methodological: any larger experiment claiming inference-time learning should be able to state analogous retained, reset, wrong-context, and reversal conditions.

## 10. Thermodynamic Mismatch Reduction as a Separate Modeling Bridge

The paper preserves the Draft 6 thermodynamic vocabulary as an analogy for local update flow:

1. A system stabilizes around a prediction, policy, plan, or context.
2. Incoming evidence introduces a property or prediction differential.
3. Local exchange or update changes the state when a gate permits it.
4. Iterated change may reduce mismatch or move the system toward another attractor.
5. Part of the adjustment may persist and affect a later episode.

| Thermodynamic language | Inference language                                                 | Status here           |
|------------------------|--------------------------------------------------------------------|-----------------------|
| Property differential  | Prediction, state, or policy mismatch                              | Structural analogy    |
| Local exchange         | Activation, routing, phase, trace, memory, or parameter adjustment | Implementation family |
| Attractor              | Stabilized percept, policy, context, or plan                       | Dynamical model       |
| Repeated dissipation   | Iterative mismatch reduction                                       | Modeling analogy      |

| Thermodynamic language | Inference language                                | Status here                          |
|------------------------|---------------------------------------------------|--------------------------------------|
| Slow persistence       | Consolidation, learned routing, or durable memory | Testable biological or AI hypothesis |

The bridge is useful because it forces the update to be local and state-bearing. It does not convert task loss into thermodynamic entropy, prove a new thermodynamic law, or make dissipation sufficient for learning. A physical energy budget and entropy-production account would require separately defined variables, units, and measurements.

## 11. Failure-Capable Experimental Program

### 11.1 AI agent test

Present repeated tasks containing a controlled, correctable mismatch. Compare fixed inference, context residue, memory residue, recurrent state, temporary-weight update, and critique-loop variants. Match token budget and computation. Evidence for the framework is a later-query difference that disappears after the specific residue is removed. The framework is weakened if improvement survives removal or is explained by extra compute.

### 11.2 Test-time adaptation test

Apply a distribution shift and measure same-shift performance over a stream. Checkpoint every candidate update surface. Compare no-update, retained update, and update-then-reset. Then reverse or switch the shift. The key outputs are  $\Delta_y$ ,  $\Delta_{\mathcal{L}}$ , persistence  $P(k)$ , and reversal harm  $H_{A \rightarrow B}$ .

### 11.3 Retrieval-memory test

Write a structured mismatch-derived memory and later query the system with relevant, irrelevant, and contradictory cues. Disable retrieval, substitute a stale record, and preserve the same context length. A valid memory residue should have a selective rather than generic effect.

### 11.4 Biological gain / timing test

Expose a circuit to a controlled mismatch and measure later response probability, gain, relative timing, or settling. Control arousal, attention, movement, stimulus history, and neuromodulatory state. A later correlation is not sufficient; a selective reset, disruption, or matched sham must alter the later response.

### 11.5 Plasticity test

Ask whether an episode-level mismatch predicts LTP / LTD, BTSP, molecular markers, synaptic efficacy, or faster settling. Test whether blocking the relevant timing window or marker removes the later effect. Competing explanations include generic salience, reward, sensory repetition, and homeostatic regulation.

### 11.6 PWD-specific test

Define a receiver-relative phase or timing mismatch before data collection. Measure the candidate PWD, the proposed fast state change, any slower consolidation, and the later response. Scramble phase while preserving rate and power where possible; reset the candidate phase / timing state; and compare a matched

non-PWD update. The proposal gains support only if the receiver-relative variable predicts a selective residue and later causal effect beyond inherited measures.

## 12. Claim Ledger

| Claim                                                | Evidence class                                            | Supported formulation                                                        |
|------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| Some inference episodes alter later inference        | Operational architecture plus executable benchmark        | Training-like status requires measured residue and controlled future effect  |
| Test-time adaptation is an engineering example       | Published AI methods [3,4]                                | Parameter or statistic updates during testing can create inspectable residue |
| Fast and slow variables coexist                      | AI and biological mechanisms [5,6,8]                      | Persistence must be measured at the relevant scale                           |
| Timing can affect plasticity                         | Primary neuroscience [7,8]                                | Timing is one candidate gate, not the only biological error mechanism        |
| Biological feedback can support learning-like update | Specific models and experiments [1,7, 12-14]              | Use “feedback-driven plasticity” and name the mechanism                      |
| PWD may act as a receiver-relative mismatch carrier  | SAN hypothesis and bounded public genealogy [15,16,19-26] | Requires direct phase / timing, residue, reset, and later-effect tests       |
| Local mismatch reduction has a thermodynamic analogy | Conceptual bridge                                         | Physical entropy and energy claims require a separate derivation             |
| SVI-style error storage is an engineering mirror     | Separate Paper 03 analysis [18]                           | It does not validate a biological mechanism                                  |

The paper’s strongest general statement is positive and testable:

Inference becomes training-like when mismatch resolution leaves a measurable state whose selective reset removes a later inference effect.

Statements about benefit, mechanism, and physical substrate remain additional questions. A harmful update can be training-like. A useful current correction can fail to leave residue. A measured residue can be causally irrelevant. The formalism separates these outcomes.

## 13. Source Ledger

| Source class                                      | Sources used here                                                                                      | What the class can establish                                                              | What it cannot establish                                                                         |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Primary neuroscience                              | Timing plasticity, BTSP, working-memory facilitation, molecular persistence, and homeostasis [6-11]    | Mechanism-specific observations and models at stated scales                               | One universal biological residue mechanism                                                       |
| Primary or official AI publications               | Test-Time Training, TENT, fast weights, feedback alignment, and predictive-coding learning [3-5,12,13] | Inspectable engineering update mechanisms and controls                                    | Validation of SAN or biological plasticity                                                       |
| Foundational predictive / prospective models      | Predictive coding, free-energy distinctions, and prospective coding [1,2,14]                           | Inherited mismatch, activity, learning, gain, and future-oriented frameworks              | Proof that every mismatch leaves residue                                                         |
| Immutable or publicly dated first-party genealogy | Public 2012 and 2017 recordings plus immutable 2022 and 2024 SAN Git revisions [15,19-26]              | Authorship, bounded content, and the public fixation date of each partial or joined stage | Independent scientific confirmation or permission to backdate the 2026 residue / reset criterion |

| Source class               | Sources used here                                                                         | What the class can establish                                                                | What it cannot establish                                               |
|----------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| First-party program papers | Prospective Learning, Building Sentient Beings, and Oscillatory State Persistence [16-18] | Development lineage and cross-paper ownership                                               | Independent evidence for their hypotheses                              |
| Executable toy benchmark   | Fixed configuration, source, tests, and byte-identical results reported in Section 9      | Internal discrimination among retained, reset, wrong-context, reversal, and null conditions | Biological evidence, general AI superiority, or real-world effect size |

Private chats and local development files were used to audit authorship and intent internally. They are not treated as public scientific evidence, are not quoted in the release candidate, and do not replace the immutable or peer-reviewed sources above.

## 14. Cross-Paper Boundaries

- **The Physics of Prospective Learning** supplies prospective state, phase-frustration framing, rapid adaptation, and slow consolidation [16]. The present paper supplies the residue / reset criterion.
- **Oscillatory State Persistence and Stable Video Infinity** owns the SVI and error-bank engineering comparison [18]. The present paper uses it only as one residue architecture.
- **Building Sentient Beings** supplies agent memory-writing and self-updating architecture [17].
- **Biological Computationalism** supplies molecular-mechanism caution and the rule that an engineering analogy does not establish a biological implementation.
- **The Cross-Disciplinary SIT Translation Guide** supplies claim labels, source-quality distinctions, and vocabulary for narrowing an overbroad statement without deleting its valid mechanism.
- **Provenance and Source-Chain Auditability** governs private-source custody, immutable locators, and the separation between authorial genealogy and independent scientific evidence.
- SAN / PWD sources supply the receiver-relative oscillatory conjecture [15,16]. They do not substitute for primary biological evidence.
- The present paper does not claim a general theory of mind, a universal learning rule, or a thermodynamic law. Its owned result is the causal boundary between transient inference and future-affecting update.

## 15. Conclusion

The difference between inference and learning cannot be settled by rhetoric or by observing that a system changed while it operated. The decisive question is whether the current episode leaves a measurable state that survives, and whether selectively resetting that state removes a later inference effect.

This paper turns that question into a formal measurement contract. It separates mismatch from update, fast state from slow consolidation, residue from causal relevance, causal relevance from benefit, and adaptation from stability. The deterministic benchmark shows why each distinction matters: retained residue improves the same-shift task, update-then-reset loses the benefit despite matched computation, wrong-context residue is harmful, and overpersistent carry fails after reversal.

The result is a shared experimental language for AI adaptation, agent memory, predictive feedback, synaptic and molecular plasticity, and the SAN / PWD research program. Those systems need not share one mechanism. They can nevertheless be compared by the same disciplined sequence:

```
measure mismatch
-> identify the update gate
```

- > locate the changed state
- > measure persistence
- > reset the candidate residue
- > test the later effect
- > test benefit and reversal separately

That sequence is what the title means. Inference is training-like when it changes what the system will do later, and when the retained cause of that change can be found.

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